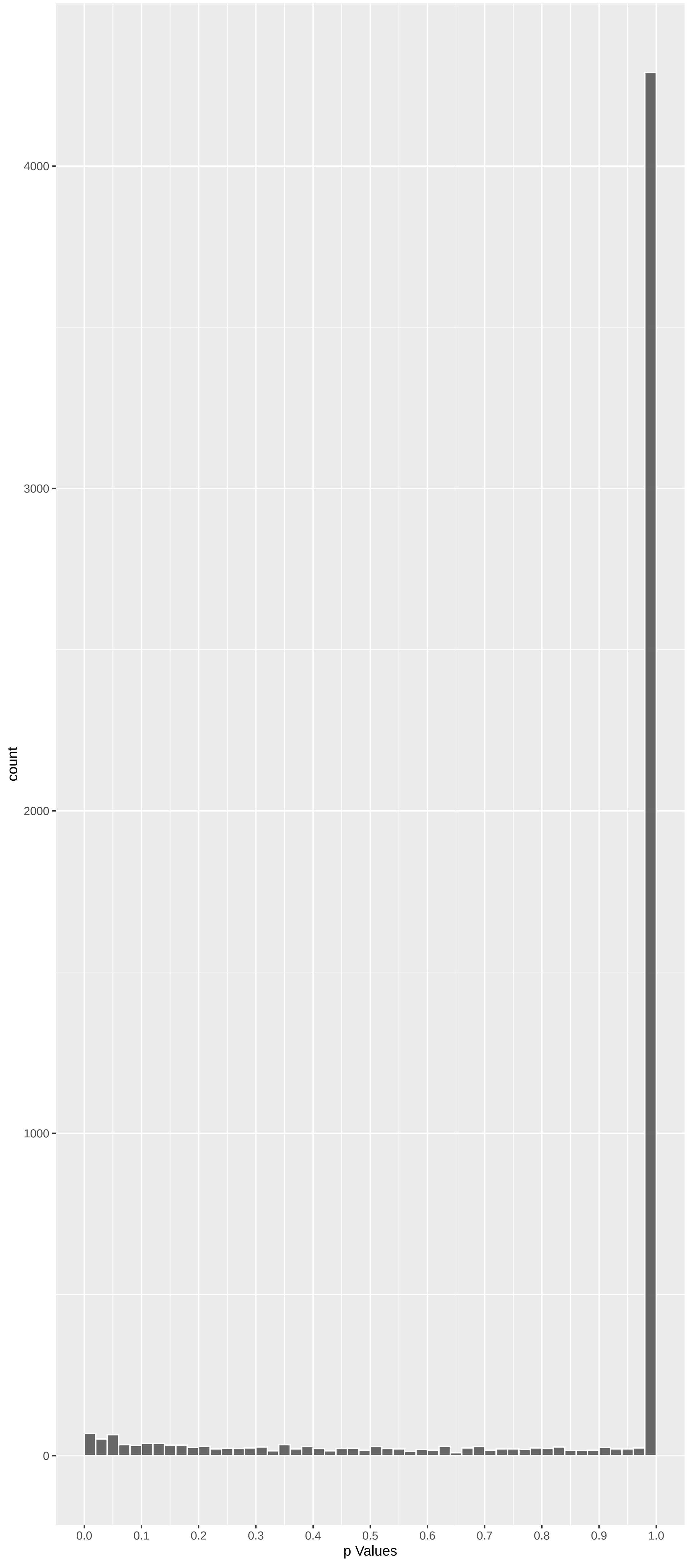
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
CEP126	−5.310615	6.555354e−07	7.142e−03	1.000e+00
STAMBPL1	−5.210847	1.127882e−06	7.142e−03	1.000e+00
RABEP1	−5.122653	1.807600e−06	7.631e−03	1.000e+00
CRTC2	−5.059567	2.521255e−06	7.983e−03	1.000e+00
DYNC2H1	−4.932156	4.879606e−06	1.236e−02	1.000e+00
MINK1	−4.808926	9.104613e−06	1.922e−02	1.000e+00
DNAH9	−4.616463	2.342016e−05	3.639e−02	1.000e+00
SLC2A10	−4.595862	2.585783e−05	3.639e−02	1.000e+00
ATP8B2	−4.608683	2.431370e−05	3.639e−02	1.000e+00
FAM205C	4.533140	3.486792e−05	4.416e−02	1.000e+00
USP45	−4.499232	4.091959e−05	4.711e−02	1.000e+00
TASOR2	−4.436220	5.493149e−05	5.798e−02	1.000e+00
SHISA5	4.412841	6.121378e−05	5.964e−02	1.000e+00
ARID4B	−4.375914	7.255480e−05	6.561e−02	1.000e+00
RSC1A1	−4.360943	7.770169e−05	6.561e−02	1.000e+00
ADGRG4	−4.293224	1.056583e−04	8.364e−02	1.000e+00
PCDH11X	−4.214733	1.500440e−04	9.049e−02	1.000e+00
COL20A1	−4.241447	1.332500e−04	9.049e−02	1.000e+00
MYRFL	−4.220776	1.460775e−04	9.049e−02	1.000e+00
TDRD12	−4.219983	1.465924e−04	9.049e−02	1.000e+00
NEK5	−4.241329	1.333203e−04	9.049e−02	1.000e+00
IQGAP3	−4.187544	1.691934e−04	9.317e−02	1.000e+00
KANK1	−4.190278	1.671680e−04	9.317e−02	1.000e+00
HIRA	4.156521	1.938780e−04	1.002e−01	1.000e+00
EGFL8	−4.151941	1.978007e−04	1.002e−01	1.000e+00

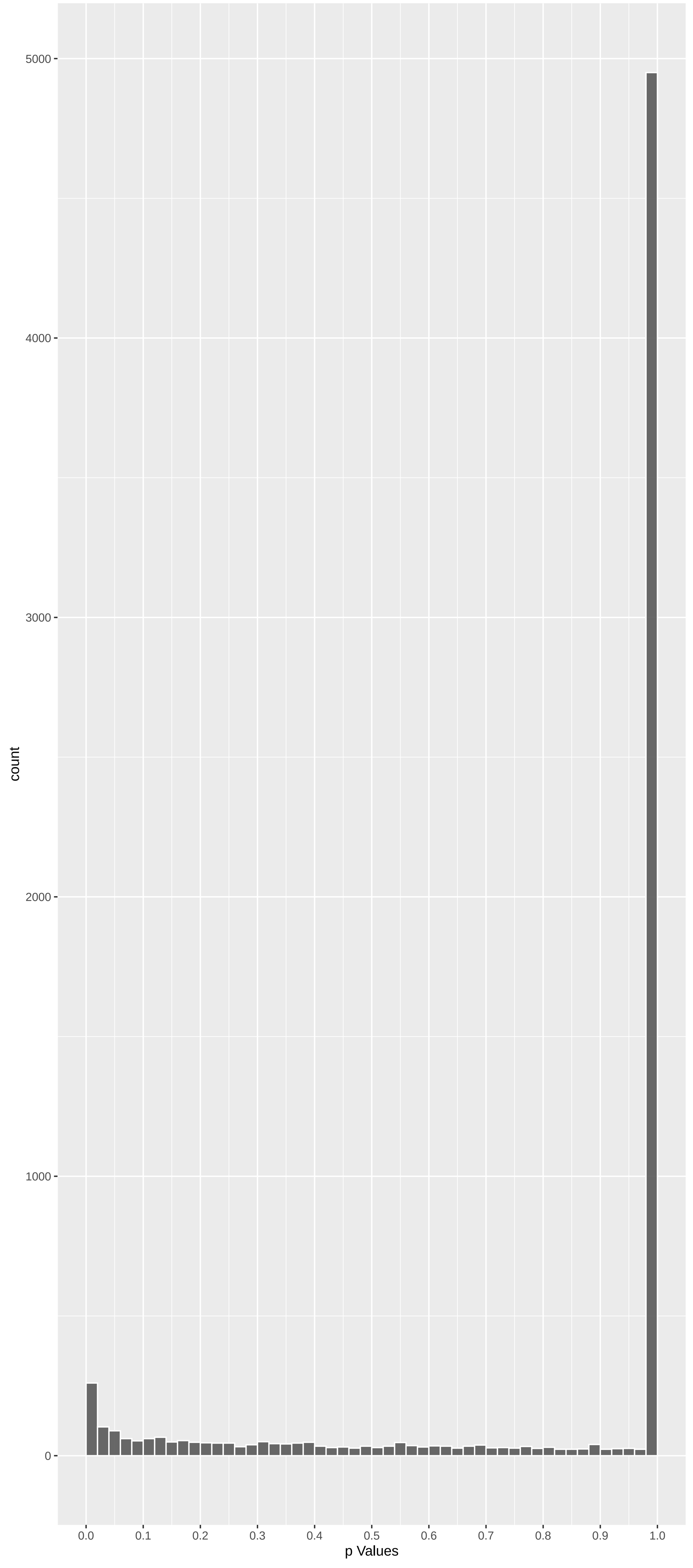
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	−0.221648797	1.00000000	1.000e+00	1.000e+00
FKBP4	1.316291509	1.00000000	1.000e+00	1.000e+00
NDUFAF7	−1.826480594	0.40666741	1.000e+00	1.000e+00
FUCA2	−1.260628516	1.00000000	1.000e+00	1.000e+00
DBNDD1	0.354085730	1.00000000	1.000e+00	1.000e+00
HS3ST1	1.367507369	1.00000000	1.000e+00	1.000e+00
SEMA3F	0.072619477	1.00000000	1.000e+00	1.000e+00
CFTR	−1.675965658	0.56246952	1.000e+00	1.000e+00
CYP51A1	2.664222724	0.04629987	1.000e+00	1.000e+00
USP28	−0.525834409	1.00000000	1.000e+00	1.000e+00
SLC7A2	0.758004532	1.00000000	1.000e+00	1.000e+00
PDK4	1.766368861	0.46400391	1.000e+00	1.000e+00
USH1C	−0.001732948	1.00000000	1.000e+00	1.000e+00
RALA	0.015058465	1.00000000	1.000e+00	1.000e+00
BAIAP2L1	0.796696746	1.00000000	1.000e+00	1.000e+00
CACNG3	0.318408436	1.00000000	1.000e+00	1.000e+00
TMEM132A	−0.034082275	1.00000000	1.000e+00	1.000e+00
DVL2	−1.009050658	1.00000000	1.000e+00	1.000e+00
RPAP3	−1.010411298	1.00000000	1.000e+00	1.000e+00
TRAPPC6A	0.435091143	1.00000000	1.000e+00	1.000e+00
SPATA20	−1.240057113	1.00000000	1.000e+00	1.000e+00
CD79B	0.323751433	1.00000000	1.000e+00	1.000e+00
RHBDD2	0.232909789	1.00000000	1.000e+00	1.000e+00
RPUSD1	1.078170471	1.00000000	1.000e+00	1.000e+00
TSR3	0.408228636	1.00000000	1.000e+00	1.000e+00

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
FAM205C	4.533140	3.486792e−05	4.416e−02	1.000e+00
SHISA5	4.412841	6.121378e−05	5.964e−02	1.000e+00
HIRA	4.156521	1.938780e−04	1.002e−01	1.000e+00
GRIFIN	4.116935	2.303662e−04	1.122e−01	1.000e+00
CLU	4.092926	2.555777e−04	1.156e−01	1.000e+00
AKR1D1	3.839323	7.402436e−04	1.700e−01	1.000e+00
IWS1	3.789110	9.071286e−04	1.922e−01	1.000e+00
SEC14L5	3.638294	1.646702e−03	2.530e−01	1.000e+00
COL1A2	3.629034	1.706903e−03	2.530e−01	1.000e+00
TLR8	3.604604	1.875773e−03	2.631e−01	1.000e+00

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
CEP126	−5.310615	6.555354e−07	7.142e−03	1.000e+00
STAMBPL1	−5.210847	1.127882e−06	7.142e−03	1.000e+00
RABEP1	−5.122653	1.807600e−06	7.631e−03	1.000e+00
CRTC2	−5.059567	2.521255e−06	7.983e−03	1.000e+00
DYNC2H1	−4.932156	4.879606e−06	1.236e−02	1.000e+00
MINK1	−4.808926	9.104613e−06	1.922e−02	1.000e+00
DNAH9	−4.616463	2.342016e−05	3.639e−02	1.000e+00
SLC2A10	−4.595862	2.585783e−05	3.639e−02	1.000e+00
ATP8B2	−4.608683	2.431370e−05	3.639e−02	1.000e+00
USP45	−4.499232	4.091959e−05	4.711e−02	1.000e+00

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cilium Assembly (GO:0060271)	−0.07760309	197	1.877e−04	5.039e−01	CEP126:1 DYNC2H1:5 KIAA0753:28 GORAB:36 TEKT1:92 DNAH5:138
Replication Fork Processing (GO:0031297)	−0.18268529	37	1.220e−04	5.039e−01	EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740 DNA2:775
DNA Repair (GO:0006281)	−0.06268710	234	1.031e−03	9.225e−01	USP45:10 XRCC5:31 ATM:60 MLH1:66 MLH3:90 EME2:124
G Protein−Coupled Glutamate Receptor Sig	0.32713099	9	6.790e−04	9.225e−01	TRPM1:87 HOMER3:113 GRM5:318 GRM8:415 GRM1:445 GRM6:491
Double−Strand Break Repair (GO:0006302)	−0.08361591	134	8.687e−04	9.225e−01	ZMYND8:23 XRCC5:31 ATM:60 EME2:124 RNF169:132 BRCA1:206
Positive Regulation Of Cellular Componen	0.11602502	70	8.064e−04	9.225e−01	CLU:5 TGFB1:22 ATP10A:56 LRP1:67 BID:143 DCTN1:267
'De Novo' AMP Biosynthetic Process (GO:0	−0.10120681	4	4.833e−01	9.602e−01	PFAS:33 PAICS:6785 ATIC:6785 ADSS1:6785 NA NA
'De Novo' Post−Translational Protein Fol	0.12126361	15	1.040e−01	9.602e−01	SDF2L1:76 GAK:122 HSPA8:475 ENTPD5:5881 PTGES3:5881 DNAJB13:5881
2−Oxoglutarate Metabolic Process (GO:000	0.05634844	13	4.819e−01	9.602e−01	OGDHL:151 D2HGDH:960 IDH1:5881 GOT2:5881 GOT1:5881 IDH2:5881
3'−UTR−mediated mRNA Destabilization (GO	0.07067574	11	4.171e−01	9.602e−01	MOV10:609 DHX36:916 RC3H1:5881 ZFP36L1:5881 ZFP36L2:5881 ZFP36:5881
3'−UTR−mediated mRNA Stabilization (GO:0	0.10615460	11	2.229e−01	9.602e−01	DAZL:107 MAPK14:886 BOLL:963 ZFP36:5881 TARDBP:5881 YBX3:5881
3'−Phosphoadenosine 5'−Phosphosulfate Me	−0.04012683	8	6.943e−01	9.602e−01	ENPP1:436 PAPSS2:1069 TPST2:6785 SULT2B1:6785 SULT1B1:6785 SULT1E1:6785
5S Class rRNA Transcription By RNA Polym	−0.05186503	5	6.880e−01	9.602e−01	GTF3C5:1509 GTF3C2:6785 GTF3C3:6785 GTF3C4:6785 GTF3C6:6785 NA
7−Methylguanosine RNA Capping (GO:000945	0.25805223	2	2.062e−01	9.602e−01	RNMT:188 CMTR2:5881 NA NA NA NA
7−Methylguanosine mRNA Capping (GO:00063	0.25805223	2	2.062e−01	9.602e−01	RNMT:188 CMTR2:5881 NA NA NA NA
ADP Transport (GO:0015866)	0.11126131	5	3.889e−01	9.602e−01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 NA
AMP Biosynthetic Process (GO:0006167)	−0.13442773	6	2.542e−01	9.602e−01	PFAS:33 APRT:864 PAICS:6785 ATIC:6785 ADSS1:6785 NUDT2:6785
AMP Metabolic Process (GO:0046033)	0.07641078	9	4.274e−01	9.602e−01	NT5E:365 AK9:1217 ADSS1:5881 NUDT2:5881 NT5C1A:5881 AMPD1:5881
ATF6−mediated Unfolded Protein Response	−0.05345055	5	6.789e−01	9.602e−01	DDIT3:1400 MBTPS1:6785 XBP1:6785 ATF6B:6785 CREBZF:6785 NA
ATP Metabolic Process (GO:0046034)	0.03454527	25	5.501e−01	9.602e−01	TGFB1:22 MYH7:163 NT5E:365 HSPA8:475 NUDT2:5881 ATP5PO:5881
ATP Synthesis Coupled Electron Transport	−0.11555397	3	4.882e−01	9.602e−01	NDUFV3:1179 NDUFB6:6785 NDUFS2:6785 NA NA NA
ATP Transport (GO:0015867)	0.02195108	9	8.196e−01	9.602e−01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 CALHM3:5881
AV Node Cell Action Potential (GO:008601	0.10753371	6	3.617e−01	9.602e−01	SCN10A:124 CACNB2:5881 SCN5A:5881 SCN4B:5881 CACNA1G:5881 TRPM4:5881
AV Node Cell To Bundle Of His Cell Commu	0.03089431	4	8.306e−01	9.602e−01	SCN5A:5881 CXADR:5881 GJC3:5881 GJA5:5881 NA NA
Arp2/3 Complex−Mediated Actin Nucleation	−0.19273439	10	3.485e−02	9.602e−01	WASHC1:500 ARPC5:662 WHAMM:1033 ARPC5L:1228 TRIM27:1940 ARPC1A:2019
B Cell Activation (GO:0042113)	0.05020335	61	1.757e−01	9.602e−01	HDAC9:183 HDAC4:242 CD86:253 CD40:389 DCAF1:400 FLT3:506
B Cell Activation Involved In Immune Res	0.03091945	12	7.108e−01	9.602e−01	MSH2:5881 GPR183:5881 CD40LG:5881 CD180:5881 LIG4:5881 RNF8:5881
B Cell Chemotaxis (GO:0035754)	−0.04914117	5	7.036e−01	9.602e−01	GAS6:1685 PIK3CD:6785 HSD3B7:6785 CXCL13:6785 CYP7B1:6785 NA
B Cell Differentiation (GO:0030183)	0.04131841	44	3.435e−01	9.602e−01	HDAC9:183 HDAC4:242 DCAF1:400 FLT3:506 TPD52:629 ZBTB7A:976
B Cell Mediated Immunity (GO:0019724)	−0.03123557	13	6.966e−01	9.602e−01	SLA2:1193 MPL:1200 IL9R:1688 CD70:6785 MSH2:6785 FCGR2B:6785
B Cell Proliferation (GO:0042100)	0.08607861	15	2.485e−01	9.602e−01	CD40:389 MEF2C:979 IL10:5881 CD70:5881 IFNE:5881 PTPRC:5881
B Cell Receptor Signaling Pathway (GO:00	0.09317267	26	1.003e−01	9.602e−01	CD38:283 CTLA4:625 VAV3:651 MEF2C:979 TEC:1073 SH2B2:1198
BMP Signaling Pathway (GO:0030509)	−0.06775707	42	1.291e−01	9.602e−01	SLC39A5:455 LEFTY1:659 RUNX2:710 GDF9:795 INHBE:813 INHBC:856
C−terminal Protein Amino Acid Modificati	−0.20084384	8	4.919e−02	9.602e−01	GPLD1:188 AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 ATG16L1:6785
C−terminal Protein Deglutamylation (GO:0	−0.30356961	4	3.550e−02	9.602e−01	AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 NA NA
C−terminal Protein Lipidation (GO:000650	−0.14089015	3	3.980e−01	9.602e−01	GPLD1:188 ATG16L1:6785 ATG7:6785 NA NA NA
C4−dicarboxylate Transport (GO:0015740)	0.03726866	10	6.833e−01	9.602e−01	SLC25A12:46 SLC25A18:569 SLC25A13:5881 SLC13A2:5881 SLC1A1:5881 UCP2:5881
CD4−positive, Alpha−Beta T Cell Activati	0.03089745	5	8.109e−01	9.602e−01	STOML2:5881 TNFSF4:5881 RSAD2:5881 NKG7:5881 HMGB1:5881 NA
CD40 Signaling Pathway (GO:0023035)	0.03937031	7	7.183e−01	9.602e−01	CD86:253 CD40LG:5881 PHB2:5881 TNIP2:5881 ITGA5:5881 RNF31:5881
CENP−A Containing Chromatin Assembly (GO	−0.05491412	5	6.707e−01	9.602e−01	HJURP:1304 MIS18A:6785 CENPI:6785 OIP5:6785 NASP:6785 NA

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.26593169	27	1.741e−06	1.128e−02	PBXIP1:32 ADAM15:55 RUSC1:113 MTX1:253 DCST2:335 THBS3:483
MIKKELSEN_ES_ICP_WITH_H3K4ME3	−0.05364367	532	2.757e−05	3.780e−02	SLC2A10:9 ZMYND8:23 PBXIP1:32 ZBTB4:42 ANXA3:48 WDCP:72
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	−0.11504674	111	2.918e−05	3.780e−02	XRCC5:31 ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419
REACTOME_MEIOTIC_RECOMBINATION	−0.22184976	31	1.926e−05	3.780e−02	ATM:60 MLH1:66 MLH3:90 BRCA1:206 RAD50:740 H2AX:792
REACTOME_DISEASES_OF_DNA_REPAIR	−0.17815757	47	2.413e−05	3.780e−02	ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTYH:211 HUS1:459
REACTOME_DEVELOPMENTAL_BIOLOGY	0.04592905	669	6.278e−05	5.816e−02	CACNA1S:17 TGFB1:22 CNTN1:26 CDK8:31 PPARA:34 DEK:54
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.05420268	466	7.010e−05	5.816e−02	COL1A2:9 TGFB1:22 IL33:24 TNFRSF4:39 RELB:47 IL1RL2:121
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	−0.19802201	33	8.314e−05	5.816e−02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	−0.11467024	98	8.981e−05	5.816e−02	XRCC5:31 ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTYH:211
WP_CILIOPATHIES	−0.09401272	148	8.217e−05	5.816e−02	DYNC2H1:5 KIAA0753:28 DCDC2:192 SDCCAG8:200 DNAH11:217 CEP164:233
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	−0.06393525	310	1.180e−04	6.371e−02	ZMYND8:23 PBXIP1:32 WDCP:72 KDM6B:180 ZNF608:216 CAVIN2:244
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	−0.14677250	58	1.121e−04	6.371e−02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HDR_THROUGH_SINGLE_STRAND_ANNEA	−0.19290778	32	1.599e−04	7.504e−02	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.06075307	330	1.622e−04	7.504e−02	SLC25A12:46 SLC20A1:49 NCKAP1:102 CHGB:117 ATP1A3:146 NPTN:192
REACTOME_INFECTIOUS_DISEASE	0.04764474	539	1.789e−04	7.722e−02	TLR8:10 ENTPD1:18 TGFB1:22 SH3GL3:38 AP1B1:44 TUBB4A:59
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME	0.12260096	77	2.032e−04	8.132e−02	HIRA:3 DGCR2:53 DEPDC5:60 ZDHHC8:130 PI4KA:231 SERPIND1:250
KAUFFMANN_DNA_REPAIR_GENES	−0.08003677	181	2.135e−04	8.132e−02	XRCC5:31 ATM:60 MLH1:66 MLH3:90 PMS2:169 BRCA1:206
REACTOME_HOMOLOGY_DIRECTED_REPAIR	−0.11198819	89	2.659e−04	9.566e−02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HOMOLOGOUS_DNA_PAIRING_AND_STRA	−0.15910278	39	5.896e−04	1.909e−01	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
REACTOME_DNA_REPAIR	−0.06671598	226	5.768e−04	1.909e−01	USP45:10 XRCC5:31 ATM:60 MLH1:66 EME2:124 PMS2:169
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.03884155	683	6.315e−04	1.947e−01	IWS1:7 COL1A2:9 TGFB1:22 SYMPK:23 CDK8:31 PPARA:34
WP_INTERACTIONS_BETWEEN_IMMUNE_CELLS_AND	0.20332916	23	7.390e−04	2.175e−01	TLR8:10 TGFB1:22 CD80:210 TGFBR2:234 CD86:253 CTLA4:625
WP_JOUBERT_SYNDROME	−0.12076775	64	8.438e−04	2.376e−01	ATM:60 CEP164:233 ARL13B:363 MKS1:438 RAB8A:681 TCTN2:693
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	−0.18559439	26	1.058e−03	2.854e−01	ATM:60 BRCA1:206 RAD50:740 DNA2:775 BRIP1:1030 WRN:1040
WINTER_HYPOXIA_METAGENE	0.06982545	176	1.443e−03	3.738e−01	TGFB1:22 PPARA:34 SLC20A1:49 ENG:51 LRP1:67 HK2:182
CONRAD_STEM_CELL	0.17512172	27	1.640e−03	4.000e−01	CD9:70 DNMT3B:140 MYC:187 PIWIL2:251 DKK3:305 RET:321
WP_T_CELL_MODULATION_IN_PANCREATIC_CANCE	0.15148380	36	1.668e−03	4.000e−01	ENTPD1:18 TGFB1:22 TNFRSF4:39 CD80:210 CD86:253 NT5E:365
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	0.07281908	156	1.745e−03	4.035e−01	IL33:24 TNFSF10:78 RRM2:89 CD163:114 BID:143 ITIH3:171
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.04240996	463	1.927e−03	4.160e−01	COL1A2:9 NFKBIE:32 HERC2:37 AP1B1:44 UBR2:50 TUBB4A:59
REACTOME_MEIOSIS	−0.12222087	54	1.907e−03	4.160e−01	ATM:60 MLH1:66 MLH3:90 BRCA1:206 STAG1:281 RAD50:740
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.06697460	180	2.003e−03	4.184e−01	TGFB1:22 TNFRSF4:39 TNFSF10:78 TGFBR2:234 TNFSF15:297 TNF:348
REACTOME_LOSS_OF_FUNCTION_OF_SMAD2_3_IN_	0.39425241	5	2.265e−03	4.314e−01	TGFB1:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA
REACTOME_SIGNALING_BY_TGF_BETA_RECEPTOR_	0.39425241	5	2.265e−03	4.314e−01	TGFB1:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA
WP_SMC1SMC3_ROLE_IN_DNA_DAMAGE_CORNELIA_	−0.31272267	8	2.192e−03	4.314e−01	ATM:60 BRCA1:206 RAD50:740 PAXIP1:1061 MRE11:1398 RAD18:2001
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05035864	312	2.349e−03	4.347e−01	COL1A2:9 TGFB1:22 IL33:24 IL1RL2:121 MYC:187 TRAF2:202
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_T	0.17889236	24	2.423e−03	4.358e−01	DEK:54 YY1:160 MYC:187 MYBL2:396 TFAP2D:529 TFAP2C:672
PID_FANCONI_PATHWAY	−0.13271960	43	2.617e−03	4.581e−01	ATM:60 BRCA1:206 FANCE:420 HUS1:459 RFC5:530 RAD50:740
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.14659116	35	2.701e−03	4.603e−01	SYMPK:23 SLU7:139 THOC5:198 LSM11:285 SLBP:409 CPSF1:443
MIKHAYLOVA_OXIDATIVE_STRESS_RESPONSE_VIA	0.42563137	4	3.196e−03	4.813e−01	OAT:694 CTSD:761 AKR1B1:1136 PGAM1:1180 NA NA
KEGG_NON_HOMOLOGOUS_END_JOINING	−0.25695973	11	3.169e−03	4.813e−01	XRCC5:31 RAD50:740 DNTT:1012 POLL:1047 XRCC4:1239 NHEJ1:1351

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hepatitis C	0.06750325	498	3.611e−07	3.526e−03	CLU:5 TLR8:10 TGFB1:22 PPARA:34 ACE:36 DEPDC5:60
Encephalomyelitis	0.08513526	254	3.471e−06	1.695e−02	CLU:5 TLR8:10 ENTPD1:18 TGFB1:22 PPARA:34 ACE:36
Dermatitis, Allergic Contact	0.13501732	83	2.202e−05	4.300e−02	PPARA:34 ACE:36 CNTN3:179 CD80:210 MRC1:224 HSD11B1:344
HIV Infections	0.05582680	511	2.045e−05	4.300e−02	TLR8:10 ENTPD1:18 TGFB1:22 GLUL:28 TRPV1:45 CD9:70
Psoriasis	0.05514324	522	2.134e−05	4.300e−02	TLR8:10 HCAR2:19 TGFB1:22 IL33:24 PPARA:34 ACE:36
Aortic Aneurysm, Abdominal	0.08580122	196	3.780e−05	6.152e−02	COL1A2:9 TGFB1:22 PPARA:34 ACE:36 FCGRT:63 LRP1:67
Seminoma	0.09992233	135	6.476e−05	9.035e−02	CLU:5 ACE:36 ENTPD6:99 AFP:118 DNMT3B:140 GGTLC3:177
Dermatitis	0.09666179	138	9.363e−05	1.046e−01	TGFB1:22 IL33:24 PPARA:34 TRPA1:40 KLK7:74 LPIN2:110
Inflammatory dermatosis	0.10557842	115	9.636e−05	1.046e−01	TGFB1:22 IL33:24 PPARA:34 TRPA1:40 LPIN2:110 PTGDS:164
Myasthenia Gravis	0.10245538	120	1.112e−04	1.086e−01	TGFB1:22 TNFRSF4:39 DNMT3B:140 CD86:253 ITGAX:274 KRT5:277
Scleroderma	0.10185243	119	1.302e−04	1.156e−01	COL1A2:9 TGFB1:22 ENG:51 RTCA:94 YY1:160 COMP:169
Dermatitis, Atopic	0.06237939	316	1.563e−04	1.174e−01	TGFB1:22 IL33:24 PPARA:34 RELB:47 KLK5:58 FCGRT:63
Eczema	0.06337804	306	1.556e−04	1.174e−01	TGFB1:22 IL33:24 PPARA:34 KLK5:58 KLK7:74 TRPM1:87
Malignant neoplasm of stomach	0.03067075	1450	1.801e−04	1.256e−01	CLU:5 COL1A2:9 TGFB1:22 SYMPK:23 IL33:24 CNTN1:26
Lymphoma, T−Cell, Cutaneous	0.08092886	177	2.182e−04	1.421e−01	TGFB1:22 TNFSF10:78 SCPEP1:128 HDAC9:183 MYC:187 TRAF2:202
Hepatocarcinogenesis	0.05865038	326	3.077e−04	1.878e−01	CLU:5 TGFB1:22 GLUL:28 PPARA:34 TRIM24:57 TNFSF10:78
Epilepsy, Temporal Lobe	0.09330872	122	3.874e−04	2.033e−01	GLUL:28 ACE:36 TRPV1:45 DEPDC5:60 TNFSF10:78 SCN2A:145
Chronic Lymphocytic Leukemia	0.03985666	708	3.954e−04	2.033e−01	ENTPD1:18 TGFB1:22 NFKBIE:32 PPARA:34 ACE:36 AQP1:42
Pemphigus Vulgaris	0.15896573	42	3.692e−04	2.033e−01	FCGRT:63 GJA8:120 DSG3:213 TNF:348 CD28:416 TNFRSF9:536
Ulcerative Colitis	0.04339224	585	4.169e−04	2.036e−01	CLU:5 TGFB1:22 IL33:24 PPARA:34 ACE:36 HERC2:37
Cataract, Pulverulent	0.44789038	5	5.237e−04	2.325e−01	GJA8:120 GJA3:238 CRYBB1:741 HSF4:954 BFSP2:1137 NA
Giant Cell Glioblastoma	0.12864985	61	5.204e−04	2.325e−01	CD9:70 TNFSF10:78 MYC:187 NOTCH3:333 BHLHE40:403 FN1:425
Tuberculosis, Pulmonary	0.09154807	117	6.489e−04	2.391e−01	TLR8:10 TGFB1:22 TRPV5:25 PDE6B:30 INTS4:69 CD9:70
Cold intolerance	0.40614217	6	5.710e−04	2.391e−01	PPARA:34 TRPA1:40 TRPV1:45 UCP1:497 IRF4:560 LPIN1:5881
Crohn Disease	0.04203590	573	7.100e−04	2.391e−01	TLR8:10 ENTPD1:18 TGFB1:22 IL33:24 PPARA:34 ACE:36
Huntington Disease	0.05048236	399	6.179e−04	2.391e−01	TGFB1:22 CNTN1:26 ACE:36 SH3GL3:38 OPN1LW:105 CD163:114
Lymphoma	0.03520338	858	6.318e−04	2.391e−01	CLU:5 TGFB1:22 IL33:24 ACE:36 FCGRT:63 TNFSF10:78
Systemic Scleroderma	0.05342320	347	7.047e−04	2.391e−01	COL1A2:9 TLR8:10 TGFB1:22 IL33:24 ACE:36 TNFRSF4:39
Virus Diseases	0.04222004	571	6.864e−04	2.391e−01	TLR8:10 TGFB1:22 IL33:24 PPARA:34 TNFRSF4:39 DEK:54
Carcinoma, Transitional Cell	0.06061514	260	8.319e−04	2.428e−01	TGFB1:22 PPARA:34 ENG:51 CD9:70 TNFSF10:78 RRM2:89
Goldenhar Syndrome	−0.32012930	9	8.834e−04	2.428e−01	TSC1:186 FRAS1:295 GRIP1:296 TBC1D20:1220 TCOF1:1582 FGF8:1651
Hepatitis B	0.04377113	506	8.858e−04	2.428e−01	TLR8:10 TGFB1:22 PPARA:34 ACE:36 DEPDC5:60 CD9:70
Herpes Simplex Infections	0.05406626	328	8.482e−04	2.428e−01	TGFB1:22 TRPV1:45 FCGRT:63 AFP:118 NCL:131 HDAC9:183
Inflammatory disorder	0.08091649	143	8.745e−04	2.428e−01	TGFB1:22 IL33:24 TNFSF10:78 LPIN2:110 CD86:253 TNF:348
Leprosy	0.10266578	88	8.951e−04	2.428e−01	CD163:114 MRC1:224 TGFBR2:234 TNFSF15:297 TNF:348 CD40:389
Liver neoplasms	0.03433840	865	8.202e−04	2.428e−01	CLU:5 CACNA1S:17 TGFB1:22 GLUL:28 PPARA:34 TNFSF10:78
Multiple Myeloma	0.03471200	827	9.199e−04	2.428e−01	TGFB1:22 ENG:51 GRK6:64 LRP1:67 CD9:70 TNFSF10:78
Other specified congenital malformation	−0.31650230	9	1.011e−03	2.531e−01	KIF7:441 BBS4:807 MKKS:925 LZTFL1:1081 BBS2:1210 BBS1:1293
Stomach Carcinoma	0.02710400	1428	1.006e−03	2.531e−01	CLU:5 COL1A2:9 TGFB1:22 IL33:24 CNTN1:26 GLUL:28
Ciliopathies	−0.07751175	150	1.099e−03	2.617e−01	DYNC2H1:5 KIAA0753:28 DNAH5:138 DCDC2:192 SDCCAG8:200 DNAH11:217

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0008875 abnormal xenobiotic pharmacok	−0.267277539	13	8.596e−04	3.381e−01	ABCC3:215 SLC47A1:222 NR1I3:522 CYP1A2:759 MTF1:1020 POR:1082
MP0010094 abnormal chromosome stability	−0.121774378	58	1.421e−03	3.381e−01	XRCC5:31 ATM:60 MCPH1:154 PMS2:169 RAG1:197 BRCA1:206
MP0008438 abnormal cutaneous collagen	−0.050946012	13	5.252e−01	9.978e−01	OGN:597 PLOD1:1042 DSE:1145 PLOD3:6785 THBS2:6785 DCN:6785
MP0002909 abnormal adrenal gland	0.070228337	16	3.315e−01	9.978e−01	CHGB:117 PTPN11:633 WNT4:1173 FOXA1:5881 SNAPIN:5881 EGLN3:5881
MP0000609 abnormal liver physiology	−0.005037642	319	7.643e−01	9.978e−01	CRTC2:4 ABCB4:103 UCN3:106 PTPN22:142 PHYH:167 UACA:173
MP0001730 embryonic growth arrest	0.040674650	137	1.047e−01	9.978e−01	HIRA:3 NCKAP1:102 EP400:159 GATAD2A:185 DCTN1:267 FLT1:372
MP0001731 abnormal postnatal growth	−0.004439600	447	7.583e−01	9.978e−01	ATM:60 HSD17B4:61 CHRNB2:68 CHRNE:74 PCDH15:99 COL10A1:153
MP0002210 abnormal sex determination	−0.027053479	260	1.431e−01	9.978e−01	ATM:60 HSD17B4:61 MLH1:66 STX2:77 PROKR2:87 MLH3:90
MP0000604 amyloidosis	0.052662907	30	3.194e−01	9.978e−01	CLU:5 RELB:47 SLC30A3:498 MSR1:548 APOE:5881 NCKAP1L:5881
MP0000598 abnormal liver morphology	0.003648248	389	8.122e−01	9.978e−01	TGFB1:22 PPARA:34 RELB:47 SLC20A1:49 LRP1:67 DNMT3B:140
MP0005334 abnormal fat pad	0.018757255	121	4.811e−01	9.978e−01	PPARA:34 ACE:36 TRPV1:45 VGF:152 TGFBR2:234 HSD11B1:344
MP0005330 cardiomyopathy	0.008579532	55	8.267e−01	9.978e−01	PPARA:34 CAV3:316 MYH11:636 MCOLN1:697 LTBP4:944 CAPN3:1102
MP0002269 muscular atrophy	0.006424505	70	8.535e−01	9.978e−01	CD86:253 CAV3:316 RET:321 VEGFA:700 LTN1:744 HNF1A:852
MP0001765 abnormal ion homeostasis	0.027604134	92	3.643e−01	9.978e−01	TRPV5:25 GLUL:28 MGAT2:260 SLC34A3:309 CALCR:407 HSPA4L:626
MP0004043 abnormal pH regulation	0.020121666	12	8.095e−01	9.978e−01	SOD2:5881 SLC4A10:5881 SLC12A1:5881 SLC4A4:5881 NOS1:5881 SCNN1B:5881
MP0008961 abnormal basal metabolism	0.017170062	15	8.181e−01	9.978e−01	VGF:152 FTO:1011 BRD2:5881 DECR1:5881 FOXA2:5881 NCOR1:5881
MP0003195 calcinosis	−0.117116816	7	2.835e−01	9.978e−01	ENPP1:436 FSHR:786 SOD2:6785 SLC12A1:6785 CASR:6785 ABCC6:6785
MP0002837 dystrophic cardiac calcinosis	−0.145199720	5	2.610e−01	9.978e−01	SLC2A4:1388 MMP9:1696 GSTZ1:6785 MYBPC3:6785 MYH6:6785 NA
MP0005369 muscle phenotype	0.038154572	167	9.409e−02	9.978e−01	COL1A2:9 CACNA1S:17 ENTPD1:18 TGFB1:22 TRIM72:35 RELB:47
MP0009250 abnormal appendicular skeleto	−0.022698576	52	5.731e−01	9.978e−01	COL10A1:153 ARSB:288 ALX4:635 RUNX2:710 PCSK5:951 DACT1:1197
MP0002998 abnormal bone remodeling	0.009134370	91	7.653e−01	9.978e−01	GPR55:75 WDR1:229 MMP14:296 FBN1:377 CALCR:407 COL1A1:417
MP0002066 abnormal motor capabilities/c	0.021856964	834	4.975e−02	9.978e−01	COL1A2:9 CACNA1S:17 TGFB1:22 CNTN1:26 TRIM72:35 HERC2:37
MP0005248 abnormal Harderian gland	−0.014875066	11	8.645e−01	9.978e−01	GLI3:1949 RARA:6785 CDKN1A:6785 CTNNB1:6785 RARB:6785 APC:6785
MP0004133 heterotaxia	−0.024078855	25	6.776e−01	9.978e−01	MKS1:438 LEFTY1:659 LEFTY2:974 POLL:1047 RPGRIP1L:1056 LAMA5:1622
MP0001529 abnormal vocalization	−0.012548488	14	8.710e−01	9.978e−01	MKKS:925 DRD2:6785 DHCR7:6785 ACHE:6785 CHRM2:6785 WFS1:6785
MP0004134 abnormal chest morphology	0.103690517	22	9.291e−02	9.978e−01	CACNA1S:17 HOXB4:85 FBN1:377 COL1A1:417 COL2A1:759 NCKAP1L:5881
MP0001845 abnormal inflammatory respons	−0.004144116	434	7.768e−01	9.978e−01	BIRC3:54 CHRNB2:68 STX2:77 ENPP2:86 NBR1:96 ABCB4:103
MP0002398 abnormal bone marrow	−0.002747017	651	8.231e−01	9.978e−01	XRCC5:31 ATM:60 CDK5RAP2:80 NBR1:96 GNRH1:116 PTPN22:142
MP0000733 abnormal muscle development	0.024371193	54	5.376e−01	9.978e−01	CACNA1S:17 MGAT2:260 MMP14:296 CDKN2A:641 MESP2:915 DLL3:1070
MP0005423 abnormal somatic nervous	0.064721703	10	4.789e−01	9.978e−01	L1CAM:84 SLC12A5:833 EGR3:5881 DSCAM:5881 CNTN2:5881 TAS1R1:5881
MP0000230 abnormal systemic arterial	−0.010607453	118	6.939e−01	9.978e−01	ARSB:288 MUS81:419 PTGIS:561 NPHS2:606 F2R:764 MKKS:925
MP0003984 embryonic growth retardation	0.004121515	249	8.269e−01	9.978e−01	HIRA:3 TGFB1:22 SLC20A1:49 ENG:51 LRP1:67 DNMT3B:140
MP0002733 abnormal thermal nociception	0.069390006	71	4.461e−02	9.978e−01	TRPA1:40 TRPV1:45 L1CAM:84 SCN10A:124 HDAC4:242 F2RL1:307
MP0005387 immune system phenotype	−0.015694731	89	6.118e−01	9.978e−01	TICAM2:452 PSMB9:540 C3AR1:627 MAP3K14:641 HCK:751 FUT7:834
MP0005386 behavior/neurological phenoty	−0.013632222	74	6.872e−01	9.978e−01	ACADVL:514 ACACB:779 SLC8A3:853 MKKS:925 CLOCK:959 GLS:1503
MP0005385 cardiovascular system phenoty	0.029229550	140	2.387e−01	9.978e−01	PPARA:34 ACE:36 MGRN1:138 ADORA2A:367 CDKN1B:413 NOTCH2:436
MP0005384 cellular phenotype	−0.024738245	341	1.290e−01	9.978e−01	DYNC2H1:5 ARID4B:12 ATM:60 COL10A1:153 ITPRIP:159 PMS2:169
MP0001544 abnormal cardiovascular syste	0.029229550	140	2.387e−01	9.978e−01	PPARA:34 ACE:36 MGRN1:138 ADORA2A:367 CDKN1B:413 NOTCH2:436
MP0005381 digestive/alimentary phenotyp	0.030125853	92	3.221e−01	9.978e−01	TGFB1:22 RELB:47 TOP3B:323 GPX1:386 COL1A1:417 FGFR1:450
MP0008995 early reproductive senescence	0.016867990	13	8.334e−01	9.978e−01	TOP3B:323 TAF4B:1040 ESR1:5881 FANCA:5881 NHLH2:5881 PFDN5:5881

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
fallopian.tube	−0.107746710	44	1.365e−02	3.686e−01	ADGRG4:14 TCF23:30 ADAMTS3:310 CFAP73:657 DTHD1:851 CDON:898
suprapubic.skin	0.060059642	174	6.868e−03	3.686e−01	CAMSAP3:13 KLK5:58 IL1RL2:121 PARD6G:155 TMPRSS11F:200 DSG3:213
cerebellar.hemisphere	0.028585017	358	6.947e−02	5.362e−01	CNTN1:26 KCNK10:27 TRIM72:35 TUBB4A:59 CHGB:117 SCN2A:145
cerebellum	0.034315040	286	4.994e−02	5.362e−01	TRIM72:35 L1CAM:84 CHGB:117 GALNT12:126 ATP1A3:146 OGDHL:151
cerebral.cortex	0.059731985	78	6.951e−02	5.362e−01	RGR:104 EIF4E1B:175 NPAS1:335 KIF1A:342 SEPTIN5:428 PDXP:432
leg.skin	0.041570125	194	4.860e−02	5.362e−01	GRIFIN:4 KLK5:58 KLK7:74 TRPM1:87 IL1RL2:121 PARD6G:155
thyroid	−0.054149628	126	3.728e−02	5.362e−01	RASEF:189 DUOX2:302 LONRF3:372 TG:457 IDO2:492 ZNF804B:502
transverse.colon	−0.044142190	128	8.710e−02	5.879e−01	UCN3:106 TRPM6:172 TRIM15:178 URAD:342 RNF186:393 ATP8B1:411
amygdala	−0.175621648	7	1.077e−01	6.464e−01	PCDH15:99 PTPRZ1:424 BCAN:879 LHFPL3:6785 NEUROD6:6785 SLC38A8:6785
atrium.auricle	−0.044528527	78	1.760e−01	8.319e−01	XIRP1:56 CHRNE:74 ALPK2:343 LMAN1L:387 SMCO1:471 CCDC141:482
substantia.nigra	0.121154174	10	1.849e−01	8.319e−01	RET:321 TH:996 CHRM5:5881 DBX2:5881 FOXB1:5881 C10orf105:5881
eye.development	0.067810458	36	1.600e−01	8.319e−01	RAX:212 SOX8:388 NOTCH2:436 ALDH3A1:807 ONECUT1:1005 NEUROD4:1029
skeletal.muscle	0.026801127	190	2.081e−01	8.635e−01	CACNA1S:17 MYLK2:20 TRIM72:35 CAVIN4:66 SLC36A2:88 MYH7:163
spleen	−0.022235344	261	2.239e−01	8.635e−01	PITPNM3:43 AKNA:146 RASGRP2:170 TTC24:191 RAG1:197 IRAK3:228
adrenal.gland	−0.033596098	93	2.656e−01	8.996e−01	CELA1:46 ABCC3:215 DNAH11:217 SLC47A1:222 SLC16A9:283 INHA:384
aorta	−0.023966365	128	3.529e−01	8.996e−01	PCDH11X:19 SUSD5:24 HMCN1:339 SYNPO:400 MYH10:463 PTGIS:561
bladder	0.039636405	18	5.608e−01	8.996e−01	PLA2G2F:586 WFDC13:846 ARL14:5881 CLEC3A:5881 CYP4B1:5881 KRT13:5881
blood	0.010061427	235	6.007e−01	8.996e−01	TLR8:10 HCAR2:19 GRK6:64 R3HDM4:199 AZU1:203 ADAM8:266
Brodmann.area.9	0.013736469	121	6.045e−01	8.996e−01	PLCB1:314 GRM5:318 NPM2:357 SYN2:606 RBFOX1:782 PPP2R2C:801
C1.spinal.cord	0.012046648	134	6.330e−01	8.996e−01	SEC14L5:8 TRPV5:25 PDE6B:30 KLK6:276 PIEZO2:508 CNDP1:533
caudate.nucleus	0.039793182	19	5.486e−01	8.996e−01	MLC1:1110 ANO3:5881 CHAT:5881 ETNPPL:5881 FAM237A:5881 GPR37L1:5881
cortex.kidney	−0.018018034	100	5.361e−01	8.996e−01	SLC28A1:63 DCDC2:192 TMEM252:254 KLHDC7A:301 RNF186:393 PKHD1:409
ectocervix	−0.048930710	22	4.275e−01	8.996e−01	COL27A1:76 RIPOR3:348 F10:846 ADRA2A:1640 MMRN1:1943 GDF7:6785
esophagus.mucosa	0.015449252	175	4.856e−01	8.996e−01	CLCA4:91 BTBD11:135 TMPRSS11F:200 DSG3:213 GLTP:216 FAT2:218
esophagus.muscularis.mucosa	−0.033600236	18	6.220e−01	8.996e−01	PRUNE2:38 RGMB:1141 COL4A6:1725 CHRM3:1918 SYNM:2070 F2RL2:6785
hypothalamus	0.039518684	45	3.603e−01	8.996e−01	MC4R:115 GABRQ:223 CALCR:407 PIRT:713 GHRH:5881 NPY2R:5881
left.ventricle	−0.022979135	57	5.498e−01	8.996e−01	XIRP1:56 SMCO1:471 ADPRHL1:639 ASB11:654 LMOD3:712 MB:852
liver	−0.008395068	294	6.270e−01	8.996e−01	CPN2:59 SLC28A1:63 ABCB4:103 SLC27A2:218 SLC16A13:220 SERPINA7:309
ovary	−0.029220170	81	3.657e−01	8.996e−01	TCF23:30 CCDC68:115 ANGPTL5:242 SUSD4:369 INHA:384 KLHDC8A:526
pancreas	−0.024948471	81	4.399e−01	8.996e−01	CTRC:120 CPB1:224 PTF1A:375 CELA2A:379 RBPJL:383 RNF186:393
peyers.patch	−0.021859603	152	3.569e−01	8.996e−01	MYRFL:17 SLC28A1:63 ANXA13:304 URAD:342 MALRD1:367 RNF186:393
putamen	0.078051904	10	3.930e−01	8.996e−01	RGS9:922 ANO3:5881 FAM237A:5881 GRM3:5881 SYNDIG1L:5881 DIPK1C:5881
sigmoid.colon	−0.045390951	39	3.278e−01	8.996e−01	CHRNA3:114 HMCN2:229 CSRP1:446 PDE3B:977 CCDC9B:1045 BMP3:1093
stomach	−0.021707233	74	5.204e−01	8.996e−01	CAPN8:243 ATP4B:258 PGC:287 CLDN23:317 ADAM28:470 AGR2:559
subcutaneous.adipose	−0.018115799	61	6.259e−01	8.996e−01	ITIH5:165 LIPE:263 GPBAR1:319 PFKFB3:679 AFAP1L1:727 PLIN2:728
tibial.artery	0.015561207	83	6.259e−01	8.996e−01	COMP:169 NOTCH3:333 SPINK13:387 BCAM:451 CCN5:709 PPP1R12B:809
uterus	−0.020901879	47	6.211e−01	8.996e−01	TCF23:30 ENPP1:436 LIPI:839 LEFTY2:974 TEX15:1234 PTGER3:1323
vagina	0.028418644	76	3.939e−01	8.996e−01	TMPRSS11F:200 DSG3:213 FAT2:218 KRT5:277 CASP14:421 PGLYRP4:422
breast	0.028067179	12	7.365e−01	9.529e−01	KRT5:277 CIDEC:558 PLIN1:658 ABCB5:5881 ADIPOQ:5881 HCAR1:5881
Brodmann.area.24	0.014099995	21	8.232e−01	9.529e−01	SLC25A18:569 STUM:663 DDN:1169 NEUROD6:5881 CNTN5:5881 CRH:5881